

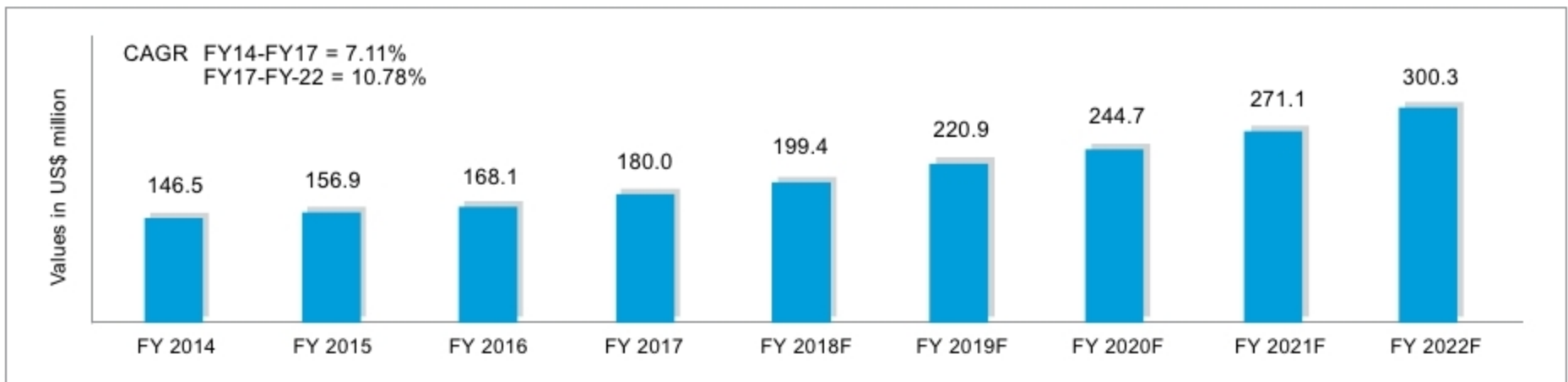


Fig. 2: Forecast of Indian T&M market size (Source: Frost & Sullivan)

## Impediments to Make in India

- Import of low-quality, cheap equipment from China
- Tough price competition
- Low customer budgets for T&M
- No major incentives or subsidies from the government for capital investments in T&M
- High GST rate of 18 per cent levied on T&M equipment; the industry has appealed to reduce the same to 12 per cent
- No major incentives for exporters of T&M equipment

technologies. Further, increasing R&D spending in the telecom sector along with growth in the cellphone market is expected to drive market growth.

Moreover, regulatory compliance, government support and constant technology developments are expected to boost the telecom testing equipment market in 2019. Considering the announcement by Telecom Engineering Centre (TEC), Department of Telecommunications (DoT), for rolling out a Mandatory Testing and Certification of Telecom Equipment (MTCTE), the Indian T&M market is expected to see huge demand for test equipment and services, which are relevant to essential requirements for safety, electromagnetic interference (EMI), electromagnetic compatibility (EMC), radio frequency (RF), specific absorption rate (SAR) and other technical requirements. Moreover, India has a vast potential for mobile manufacturing, too. This, in turn, creates significant demand for T&M equipment and solutions related to mobile handset manufacturing.

The trend of automotive companies switching to automated testing is also driving up demand for T&M equipment. Moreover, increasing electronic content in vehicles for infotainment, connectivity and safety purposes, along with growing acceptance of green mobility (through e-vehicles)

will also fuel the growth of the T&M sector. Besides, increase in R&D activities across the electronics industry value chain is expected to further drive up demand.

Demand for industrial automation across India has been growing at a rapid rate and is expected to grow further in the years to come. With growing demand, power generation, transmission and distribution companies will have to invest in reliable, accurate and efficient equipment to protect their assets from failure. Therefore growth prospects of T&M equipment related to the industrial electronics sector are quite promising.

India is an emerging solar hub of the world. Rise in the manufacturing of solar or photovoltaic cells requires high-performance test equipment for solar cell and module testing, which include power supplies, digital multi-meters and electronic loads.

The strong government push to make in India in the aerospace and defence electronics sector indicates the bright future of the defence manufacturing ecosystem in India and related opportunities for the T&M segment as well.

Demand for oscilloscopes, spectrum analysers, RF communication testers and EMC solutions will grow at a high rate from industrial and automotive, aerospace and defence, and communications segments. According to the Frost & Sullivan report (Fig. 3):

- The oscilloscopes market is estimated to grow at a CAGR of 13.3 per cent from FY17 to FY22 due to increasing demand from all end-user segments.
- Demand for spectrum analysers will increase with the advent of new telecom technologies.

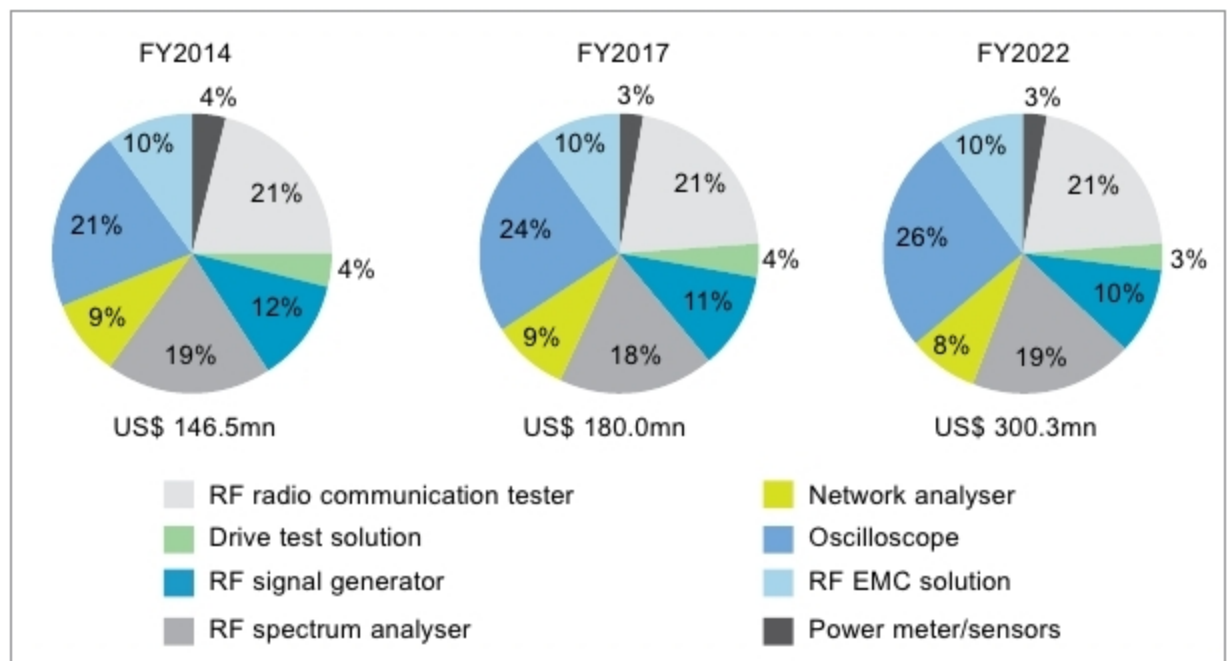


Fig. 3: Forecast of Indian T&M market by product type (Source: Frost & Sullivan)